Degly Sebastián Pava Pava

deglypavapava@gmail.com https://deglypava.com IMLEX and Systems and Computer Engineer

From: August 2021
To: September 2023
(Currently in Japan)

Erasmus Mundus Japan – Master of Science in Imaging and Light in Extended Reality (IMLEX)

- *Quality Board Member and Communications Representative*
- *Triple Degree Program (Msc. Optics, Image, Computer Vision, Msc. Engineering, Msc. Computer Science)*
- *Toyohashi University of Technology (Japan), KU Leuven (Belgium), University of Eastern Finland (Finland), University Jean Monnet Saint-Etienne (France)*

Tech: Photonics | Realtime 3D Rendering | Computational Imaging | CUDA | OpenMP | Computer Graphics | Physical Optics | Machine Learning | ARKit | ARCore | VR/AR.

WORK EXPERIENCE

Company Owner

Omni Applications L.L.C. (Florida, U.S.A) – Lucid Dreams Ltd. (Bogotá, Colombia)

Tech: Project Management | ThreeJS | iOS | XCode | Unreal | SparkAR | 360 | LensStudio | Blender | Jira | Slack

From: January 2018
To: August 2021

Lead XR Developer | Hackspace for Innovation and Visualization in Education

Faculty of Medicine, The University of British Columbia, Vancouver, B.C. (Canada)

- As part of THE HIVE Research Laboratory, contributed to The HoloBrain and MR Manikin / OCS projects by creating multiple 3D experiences and integrating them into a mixed reality environment using concepts of 3D visual computing. Utilized Unity 3D Engine to develop the experiences and released them through various devices, including Oculus Rift, HTC VIVE, and Extended Reality devices.

Tech: Unity | Microsoft HoloLens 1&2 | OpenXR | Mixed Reality Toolkit (MRTK) | Vuforia | Blender | Neuroanatomy Azure | Cloud Computing | Dependency Injection | Reactive Programming

From: February 2017
To: September 2019

Lead Developer and Engineer at Pontifical Xavierian University

Department of Industrial and Systems Engineering at Pontifical Xavierian University, Bogotá (Colombia)

- Created a 3D Material Requirements Planning (MRP) simulation system using NetLogo to optimize production planning, scheduling, and inventory control. Currently used in production classes to train students in MRP and scheduling concepts.

Tech: NetLogo | Anylogic | Linear Algebra | Microsoft Office Suite | Adobe Suite | Research Techniques

EDUCATION AND TRAINING

Certifications: *VR / MR / XR Developer* · *Meta Spark Creator AR* · *Meta AR Developer* · *AR & Video Streaming Services*

Academic Recognitions:

Bachelor of Systems and Computer Engineering

- *Special Admission Program, best high school graduates of Colombia Scholarship at Universidad Nacional de Colombia*
- *IronHacks – Purdue University Fellowship*
- *Emerging Leaders of the Americas Program*

Master of Imaging and Light in Extended Realities

- *IMLEX Full Tuition Award*
- *ERASMUS Mundus Japan Scholarship*
- *COLFUTURO PCB Scholarship*
- *Manutech SLEIGHT Certificate Scolarship*
- *BRMI Academic Mobility Award*

Trained in multiple research groups:

Visual Perception and Cognition Laboratory	*Toyohashi University of Technology - Japan*
Hackspace for Innovation and Visualization in Education	*The University of British Columbia - Canada*
Research Center for Open Digital Innovation	*Purdue University – United States*
Grupo de investigación en procesos químicos y bioquímicos	Universidad Nacional de Colombia

From August 2021
To: September 2023

- Engineered a VR research experience, leveraging eye tracking technology and drawing from human psychology and HCI principles to analyze pupil responses in an immersive virtual environment.

From: January 2019
To: April 2021

- Designed and developed advanced neuroanatomy applications using MRI (Magnetic Resonance Imaging) and VR (Virtual Reality) technologies for medical research and education.

From: June 2018
To: December 2018

- Contributed to organizing the Gold IronHacks Purdue-UNAL Hackathon 2018. Also served as a judge, assessing participants' websites for the best implementation of Open Data based on specific attributes.

Digital skills	Extended Realities	Programming	Optics / Physics	Research and Tools
	Unity | Blender | ThreeJS	C# | Python | C++	Photonics | Colorimetry	ImageJ3D | Healthcare IT
	AWS | Android | UX design	JavaScript | NetLogo | Metal	Eye-Tracking | Robotics	TensorFlow | Photon Realtime
	Vuforia | XR SDKs | Arduino	Git | Swift | WebXR	Spectral Imaging	Mendeley | Adobe Suite

PERSONAL SKILLS

Spanish Native proficiency

English Bilingual - IELTS (8.0) and iBT TOEFL (103 points)

French Limited Working Proficiency

Japanese Furigana / Basic Speaking and Writing

AWARDS AND RESEARCH PROJECTS

- ***HoloBrain 2.0 – ELAP Scholarship - THE HIVE at The University of British Columbia***
  A mixed reality brain simulation to allow educators to teach the anatomy of the brain with a holographic model, using 3D reconstructions of basal ganglia nuclei which were obtained from MRI scans. Developed a volumetric rendering that could behave as a 3D Mesh and be interactable with hand gestures so the user could cut the brain in real time to have a better idea of its parts and how it was internally organized. A paper is being written on how 3D Visualizations are better than 2D materials.

- ***MR. Manikin / Online Clinical Simulation – VIRS Research Fellowship - THE HIVE at The University of British Columbia***
  A mobile AR application that would enable medical students to interact with a critical care patient. Developed a 3D model of the patient, used Vuforia Engine to place it on the real world and Photon Networking within Unity 3D to send instructions from the Teacher Facilitator (iPad or PC) to the student interacting with the patient (Android or iOS). The application is currently in beta, soon to be published on the App Stores and its being used on multiple medical research projects.

- ***Energy Modelling Tool – RCODI's Research Fellowship - Purdue's Research Center for Open Digital Innovation***
  HTML and AWS research-based software focused on understanding what civil engineers and architects need to find the best way to build an energy efficient building by using data obtained from EnergyPlus processing, Open Data and MATLAB Cloud Computing. All of these tools were used for cost calculation and component selection.

- ***Winner of Purdue-UNAL Gold IronHacks 2017***
  Developed a website with a mashup that utilized Open Data to help students rent houses near the University of Illinois, Chicago. Open Data was used to create interactive visualizations to give insights and support optimal decisions. The project was developed through Git and judged by a team of experts from Purdue University. Later on I had the chance to join RCODI and help with the research project that analysed how the students who participated, used the Open Data provided in addition to visualizations such as D3.js.

- ***Lead developer, Simulation of TransMilenio – Full Scholarship at Universidad Nacional de Colombia***
  Developed a NetLogo simulation of Bogota's transport service TransMilenio to try to optimize how buses should be deployed to improve the quality of service. The software simulated how the system works, showing its flaws and delays. I also propose an algorithm to improve its quality. The project required multiple measurement visits to the system and optimizations to the model. In the end we were able to propose a different approach to how the buses are deployed in order to improve the bad service provided.